network gradient bioprinting collagen migration traction hydrogel collagen fluorescent crosslink polymer scaffold coating confocal rheology confocal bioprinting rheology migration scaffold hydrogel rheology network coating displacement density viscoelastic spheroid bioprinting modulus bead elastic microscopy bioprinting bioprinting elastic force matrix fluorescent hydrogel rheology density polymer substrate microscopy porosity traction rheology migration substrate confocal tissue field diffusion stiffness strain collagen cellular elastic porosity fibroblast stiffness rheology crosslink migration stress substrate stiffness density field crosslink microscopy crosslink porosity network stiffness fluorescent granule field matrix network displacement traction strain bead bioprinting stress viscoelastic adhesion scaffold modulus cellular coating migration cellular stress modulus granule bioprinting matrix cellular strain network collagen viscoelastic tissue viscoelastic viscoelastic hydrogel microscopy field elastic traction hydrogel elastic density density gradient substrate crosslink rheology substrate microscopy migration coating collagen matrix bioprinting modulus stress stiffness fibroblast gradient granule rheology polymer force traction adhesion adhesion matrix strain tissue modulus hydrogel migration network rheology spheroid substrate bioprinting modulus elastic tissue collagen tissue crosslink hydrogel spheroid scaffold gradient confocal network force migration force tissue modulus elastic polymer granule coating gradient strain density hydrogel crosslink migration crosslink network fibroblast density bioprinting density migration granule field field hydrogel fibroblast scaffold network field hydrogel diffusion viscoelastic gradient fibroblast density stress stiffness field elastic migration rheology gradient bioprinting stress polymer field scaffold bioprinting density stiffness gradient strain migration rheology density crosslink hydrogel displacement cellular microscopy rheology displacement viscoelastic modulus displacement field displacement crosslink displacement displacement viscoelastic fluorescent gradient fluorescent fluorescent crosslink confocal fluorescent network density migration gradient rheology elastic modulus bioprinting hydrogel cellular strain traction collagen coating migration bead hydrogel elastic confocal confocal field hydrogel collagen stiffness microscopy fibroblast traction hydrogel collagen gradient microscopy network traction fibroblast network field stiffness fibroblast microscopy density modulus stiffness strain cellular cellular bead bioprinting substrate strain microscopy elastic porosity network hydrogel field adhesion matrix migration elastic force force diffusion adhesion cellular diffusion fibroblast field fluorescent crosslink spheroid modulus hydrogel collagen collagen network fibroblast polymer viscoelastic adhesion cellular collagen tissue force tissue field viscoelastic strain gradient rheology gradient density adhesion stress displacement spheroid collagen displacement rheology crosslink porosity crosslink field polymer density scaffold collagen rheology spheroid modulus field scaffold traction traction gradient crosslink field stiffness bioprinting coating substrate granule hydrogel adhesion tissue collagen strain elastic gradient crosslink traction gradient stiffness elastic fibroblast elastic spheroid cellular fluorescent spheroid gradient stiffness gradient strain gradient traction substrate fibroblast force traction network crosslink gradient spheroid gradient spheroid diffusion adhesion stress collagen displacement coating tissue field bead confocal field field scaffold rheology tissue hydrogel rheology tissue matrix collagen elastic gradient confocal gradient spheroid bioprinting polymer porosity density adhesion substrate crosslink traction hydrogel diffusion stress polymer substrate scaffold bead migration modulus granule traction tissue polymer fibroblast fibroblast coating granule coating collagen stress stress granule traction polymer matrix fibroblast hydrogel viscoelastic fluorescent granule elastic density crosslink migration spheroid viscoelastic modulus strain crosslink substrate microscopy substrate stiffness strain force bead viscoelastic migration coating field microscopy rheology fibroblast cellular network force scaffold stress bead stress density displacement coating scaffold polymer polymer migration modulus rheology substrate fibroblast fluorescent hydrogel strain stiffness crosslink diffusion rheology confocal gradient collagen cellular spheroid elastic network density matrix cellular diffusion fluorescent network strain migration matrix collagen displacement microscopy tissue fibroblast substrate fluorescent scaffold coating fibroblast scaffold stiffness stress modulus modulus bioprinting tissue viscoelastic granule rheology matrix diffusion migration coating granule crosslink fibroblast gradient spheroid elastic adhesion modulus force tissue displacement hydrogel diffusion coating migration fluorescent microscopy force matrix stiffness bioprinting fibroblast rheology cellular rheology displacement force spheroid confocal fluorescent porosity force crosslink crosslink scaffold tissue density spheroid cellular modulus crosslink cellular matrix field porosity scaffold modulus tissue diffusion adhesion stress cellular stiffness stiffness density force scaffold displacement microscopy fluorescent diffusion spheroid tissue field migration spheroid diffusion microscopy microscopy field microscopy adhesion scaffold elastic diffusion modulus